Mohsen Fayyaz

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Education

University of California, Los Angeles — *Los Angeles, CA*
Ph.D. in Computer Science — *Sep. 2023 - Present*
- advisor: **Nanyun (Violet) Peng**
- GPA: 3.95/4

University of Tehran — *Tehran, Iran*
Master's in Artificial Intelligence, Computer Engineering — *Sep. 2021 - Aug. 2023*
- advisor: Yadollah Yaghoobzadeh
- GPA: 20/20 (4/4)

University of Tehran — *Tehran, Iran*
B.Sc. in Computer Engineering — *Sep. 2017 - Sep. 2021*
- Ranked 3rd among 103 Computer Engineering students
- GPA: 19.08/20

Research Interests

- Natural Language Processing
- Actionable Interpretability

Publications

14. Diffusion Language Models Change Speed Across Tasks and Languages — *arXiv preprint*
- **M Fayyaz**, A Modarressi, D Israel, L Bandarkar, N Peng

13. From Routing Signals to Selective Review: Visual regrounding in MoE VLMs — *arXiv preprint*
- H Guo, **M Fayyaz**, N Peng

12. Multilinguality in Hybrid Attention LLMs — *arXiv preprint*
- L Bandarkar, J Hu, C Yang, **M Fayyaz**, N Peng

11. Multilingual Routing in Mixture-of-Experts — *ICLR 2026*
- L Bandarkar, C Yang, **M Fayyaz**, J Hu, N Peng

10. Steering MoE LLMs via Expert (De)Activation — *ICLR 2026*
- **M Fayyaz**, A Modarressi, H Deilamsalehy, F Dernoncourt, R Rossi, T Bui, H Schütze, N Peng

9. Collapse of Dense Retrievers: Short, Early, and Literal Biases Outranking Factual Evidence — *ACL 2025*
- **M Fayyaz**, A Modarressi, H Schuetze, N Peng

8. MRAG-Bench: Vision-Centric Evaluation for Retrieval-Augmented Multimodal Models — *ICLR 2025*
- W Hu, J Gu, Z Dou, **M Fayyaz**, P Lu, K Chang, N Peng

7. Evaluating human alignment and model faithfulness of llm rationale — *arXiv preprint*
- **M Fayyaz**, F Yin, J Sun, N Peng

6. SkipPLUS: Skip the First Few Layers to Better Explain Vision Transformers — *CVPR Workshops 2024*
- F Mehri, **M Fayyaz**, MS Baghshah, MT Pilehvar

5. DecompX: Explaining Transformers Decisions by Propagating Token Decomposition — *ACL 2023*
- A Modarressi*, **M Fayyaz***, E Aghazadeh, Y Yaghoobzadeh, MT Pilehvar
- (* equal contribution)

4. BERT on a Data Diet: Finding Important Examples by Gradient-Based Pruning — *ENLSP @ NeurIPS 2022*
- **M Fayyaz***, E Aghazadeh*, A Modarressi*, MT Pilehvar, Y Yaghoobzadeh, SE Kahou
- (* equal contribution)

3. GlobEnc: Quantifying Global Token Attribution by Incorporating the Whole Encoder Layer in Transformers — *NAACL 2022*

- A Modarressi*, **M Fayyaz***, Y Yaghoobzadeh, MT Pilehvar
- (* equal contribution)

2. Metaphors in Pre-Trained Language Models: Probing and Generalization Across Datasets and Languages — *ACL 2022*

- E Aghazadeh*, **M Fayyaz***, Y Yaghoobzadeh
- (* equal contribution)

1. Not All Models Localize Linguistic Knowledge in the Same Place: A Layer-wise Probing on BERToids' Representations — *BlackboxNLP @ EMNLP 2021*

- **M Fayyaz***, E Aghazadeh*, A Modarressi, H Mohebbi, MT Pilehvar
- (* equal contribution)

Work Experience

Research Intern — *Seattle, Washington, United States*

Adobe — *Jun. 2025 - Sep. 2025*

- Steering MoE LLMs via Expert (De)Activation

Data Scientist — *Tehran, Iran*

Divar — *Jun. 2023 - Aug. 2023*

- Developed NLP services including various ML tasks and models at large scale.

Data Scientist — *Tehran, Iran*

Sotoon — *Jul. 2022 - Jun. 2023*

- Developed NLP/AutoML services including various ML tasks and models at large scale.

Machine Learning Engineer — *Tehran, Iran*

Tapsell — *Sep. 2021 - Jul. 2022*

- Developed NLP engines and ML models for enhancing targeted advertising.

Research Experience

Graduate Student Researcher at University of California, Los Angeles — *Los Angeles, CA*

Under supervision of Prof. Nanyun (Violet) Peng — *Fall 2023 - Present*

- Ph.D. Researcher at PLUS Lab

Remote Collaborator at École de technologie supérieure — *Montreal, Canada*

Under supervision of Prof. Samira Ebrahimi and Prof. Yadollah Yaghoobzadeh — *Spring 2022*

- BERT on a data diet: Finding important examples by gradient-based pruning (ENLSP @ NeurIPS 2022)

Graduate Research Assistant at University of Tehran — *Tehran, Iran*

Under supervision of Prof. Yadollah Yaghoobzadeh and Prof. Mohammad Taher Pilehvar — *2021 - 2023*

- Quantifying global token attribution by incorporating the whole encoder layer in Transformers. (NAACL 2022)
- Metaphors in pre-trained language models: Probing and generalization across datasets and languages. (ACL 2022)

NLP Research Intern at Tehran Institute for Advanced Studies (TeIAS) — *Tehran, Iran*

Under supervision of Prof. Mohammad Taher Pilehvar — *Summer 2021*

- Layer-wise probing on BERT, XLNet, and ELECTRA representations for linguistic knowledge. (BlackboxNLP @ EMNLP2021)

Undergraduate Research Assistant at University of Tehran — *Tehran, Iran*

Under supervision of Prof. Azadeh Shakery — *Spring 2021*

- Investigating toxicity detection knowledge in contextual language models and possible biases.

Teaching

Teaching Assistantship

2025 **CS162 - Natural Language Processing**, Prof. Nanyun (Violet) Peng *UCLA*
2024 **CS263 - Natural Language Processing**, Prof. Nanyun (Violet) Peng *UCLA*
2020-2022 **Artificial Intelligence**, Prof. Yaghoobzadeh, Dr. Fadaei *University of Tehran*
2019-2020 **Formal Languages and Automata Theory**, Prof. Hojjat *University of Tehran*
2019-2020 **Advanced Programming**, Prof. Khosravi *University of Tehran*
2019 **Digital Logic Design**, Prof. Navabi *University of Tehran*